What is work? (M1.1)

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What Is Work? Why Definition Matters

- “Work” feels obvious in daily life
- Engineering and design require a definition of “work” that supports:
  - Analysis
  - Measurement
  - Redesign
- Different disciplines emphasize different aspects of work
- Different perspectives function like different zoom lenses:
  - At different angles
  - From micro actions to full socio-technical systems

What Is Work? Why Definition Matters

- Levels of describing work:
  - Action / motion (e.g., reach, click)
  - Task (e.g., assemble a part, enter an order)
  - Process (interrelated activities transforming input to output)
  - Work system (organized socio-technical system in which processes occur)

Economic Perspective on Work

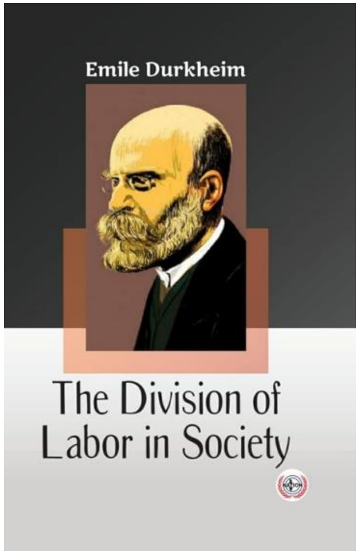
- Labor = core input to production
- Productivity = output per input
- Explains:
  - Living standards
  - Long-run economic growth

Economic Perspective on Work

- Limitations of narrow economic view:
  - Work ≠ only paid employment
  - Unpaid work matters:
    - Care work
    - Household work
    - Coordination work
- OECD "third-person criterion":
  - If someone else could be paid to do it, it qualifies as work
  - Enables inclusion of operationally relevant unpaid work

Sociological Perspective on Work

- Work is embedded in:
  - Division of labor
  - Institutions
    - Roles
    - Rules
    - Authority
    - Norms
- Durkheim:
  - Specialization (division of labor) is not only technical
  - It reshapes social coordination and interdependence



Sociological Perspective on Work

- Implication for work system design:
  - Technology changes:
    - How tasks are performed, and, also
    - Who decides
    - Who coordinates
    - Who is responsible
    - How performance is judged

Psychological Perspective on Work

- Distinction:
  - Outcomes (e.g., sales)
  - Performance (goal-relevant behaviors)
- Performance:
  - Observable behaviors
  - Measurable in terms of contribution
  - More controllable than outcomes via system design

Psychological Perspective on Work

- Job design research:
  - Autonomy
  - Feedback
  - Task meaning

    → Influence motivation, job satisfaction, engagement, and performance

Psychological Perspective on Work

- Autonomy in Job Design
  - Definition: Degree of freedom and discretion in how work is performed
  - Key questions
    - Can the worker decide methods or sequence?
    - Can they control pacing?
  - Example
    - Low: strict step-by-step instructions
    - High: choice of tools, sequence, approach
  - Psychological impact
    - Increases responsibility and intrinsic motivation
  - Work design implication
    - Provide structured guidance
    - Allow intelligent judgment where appropriate

Psychological Perspective on Work

- Feedback in Job Design
  - Definition: Clear information about performance effectiveness
  - Types
    - From the job (e.g., machine/error display)
    - From others (e.g., supervisor/customer)
  - Example
    - Low: no visibility into quality
    - High: real-time performance dashboard
  - Psychological impact
    - Reduces uncertainty
    - Supports learning and motivation
  - Work design implication
    - Ensure information visibility and clear performance signals

Psychological Perspective on Work

- Task Meaning in Job Design
  - Meaningfulness arises from
    - Task significance (impact on others)
    - Task identity (completing a whole task)
    - Skill variety
  - Definition: Perceived importance and value of the task
  - Example
    - Low: repetitive fragment with no context
    - High: understanding contribution to final outcome
  - Psychological impact
    - Increases intrinsic motivation and persistence
    - Reduces burnout
  - Work design implication
    - Design tasks so workers:
      - See outcomes
      - Understand impact
      - Perform coherent (not overly fragmented) work

Psychological Perspective on Work

- Implication for work system design:
  - Work redesign is not only about speed
  - Also about:
    - Information
    - Responsibility
    - Feedback
    - Meaning

Human Factors / Ergonomics Perspective

- Work = human–system interaction
- Goals (IEA):
  - Improve human well-being
  - Improve overall system performance

Human Factors / Ergonomics Perspective

- Balanced Socio-Technical Design (ISO 6385)
  - Balanced approach = joint optimization of:
    - Human requirements
      - Physical and cognitive limits
      - Fatigue, workload, learning
    - Social/organizational requirements
      - Roles and responsibilities
      - Authority and coordination
      - Incentives and communication
    - Technical requirements
      - Tools, machines, software
      - Process flow and standards
      - Layout and automation

  - Key principle:
    - Do not optimize one dimension (e.g., speed) at the expense of others
    - System performance emerges from interaction among all three

Human Factors / Ergonomics Perspective

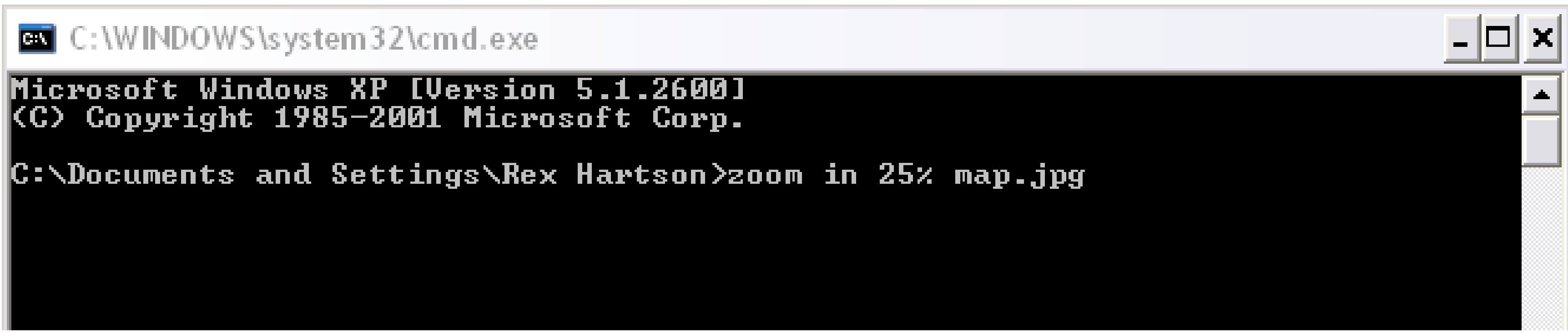
- Key message:
  - Improvement ≠ squeezing people
  - Improvement = reducing unnecessary load and variability
  - Enhancing:
    - Safety
    - Quality
    - Experience

Human Factors / Ergonomics Perspective

- What Is “Unnecessary Load”?
  - Load types
    - Physical load (force, posture, repetition)
    - Cognitive load (memory, attention, decision effort)
    - Emotional load (stress, conflict)
  - Unnecessary load = strain that does not create value
  - Examples:
    - Searching for tools
    - Re-entering data
    - Monitoring poorly designed displays
    - Remembering information that could be visible
  - Goal:
    - Remove avoidable strain
    - Preserve effort that contributes to value

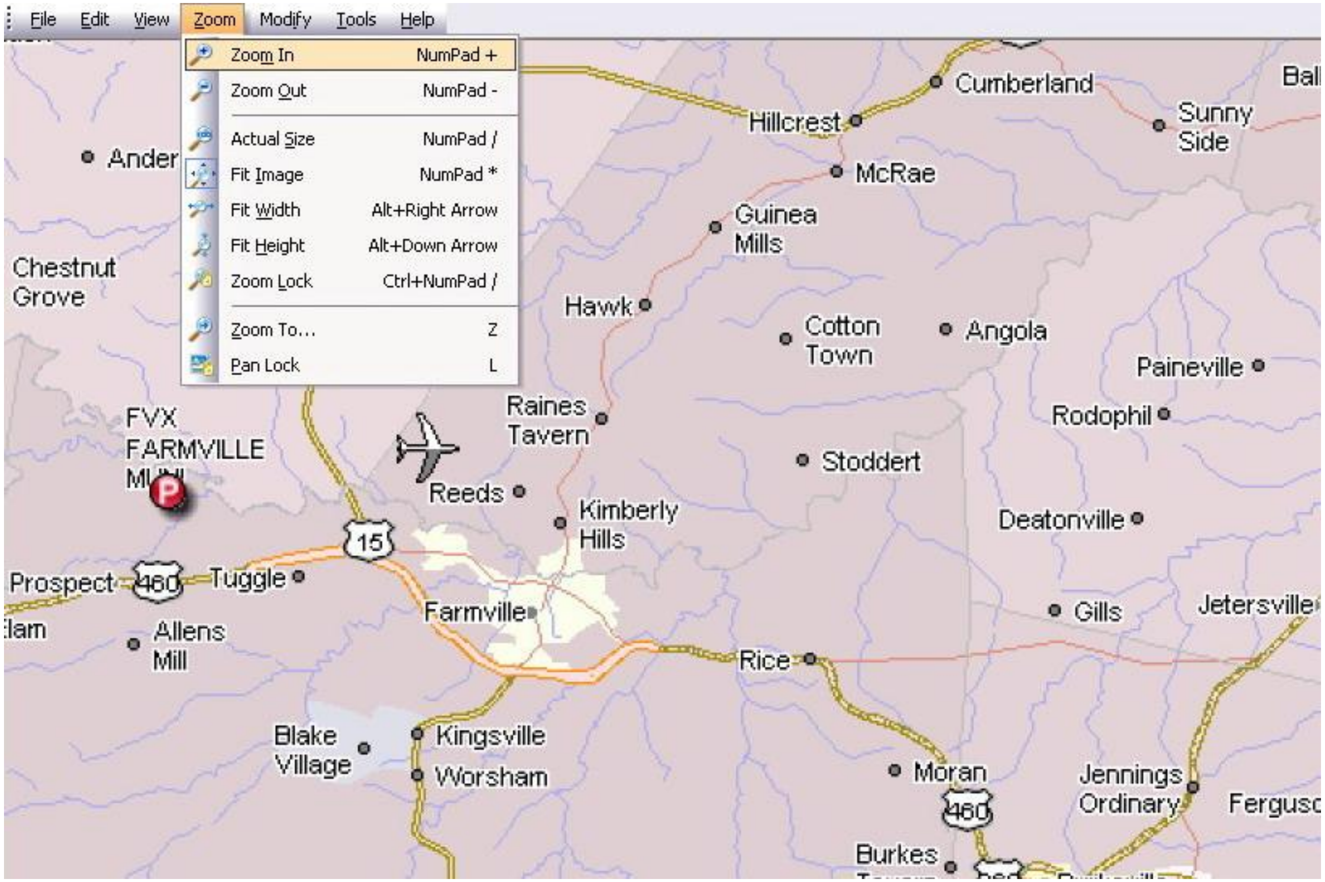
Reducing load: zooming in on map image (1)

Command, language input



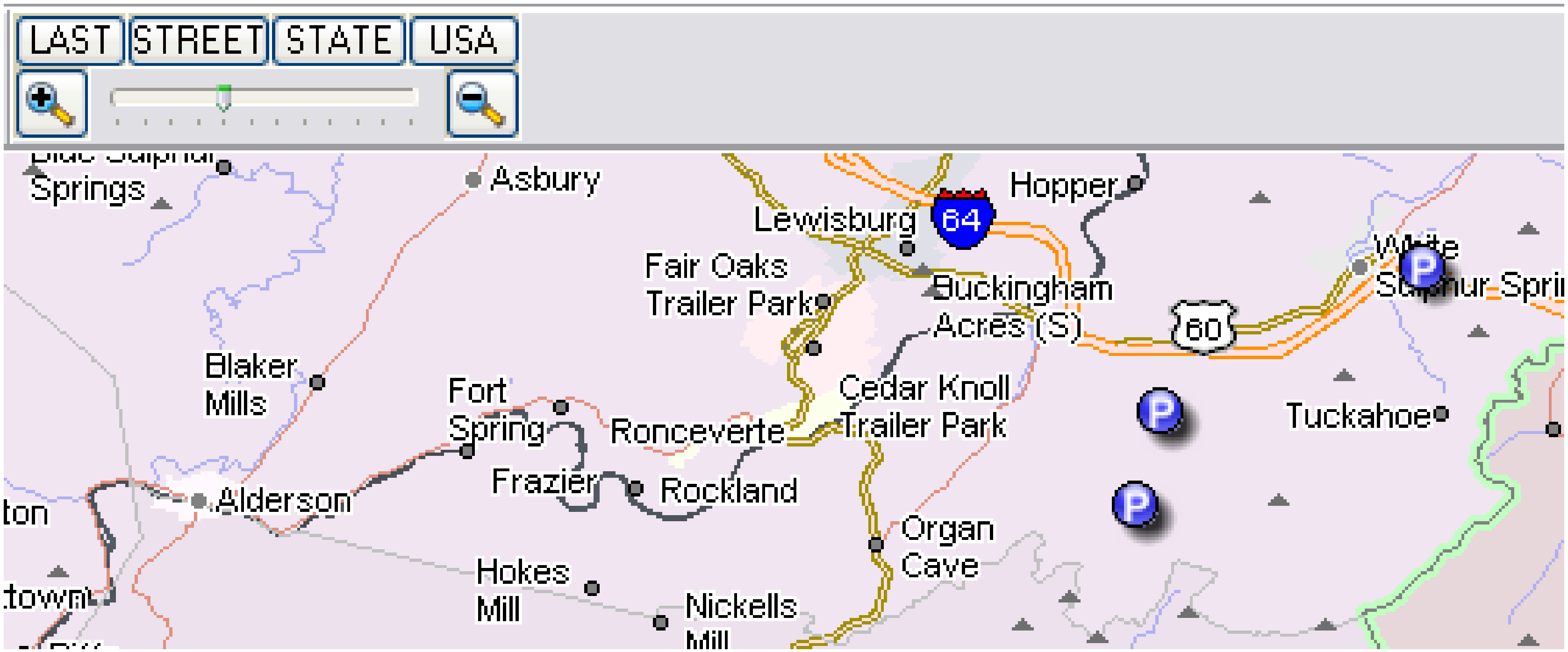
Reducing load: zooming in on map image (2)

Command, via a pull-down menu



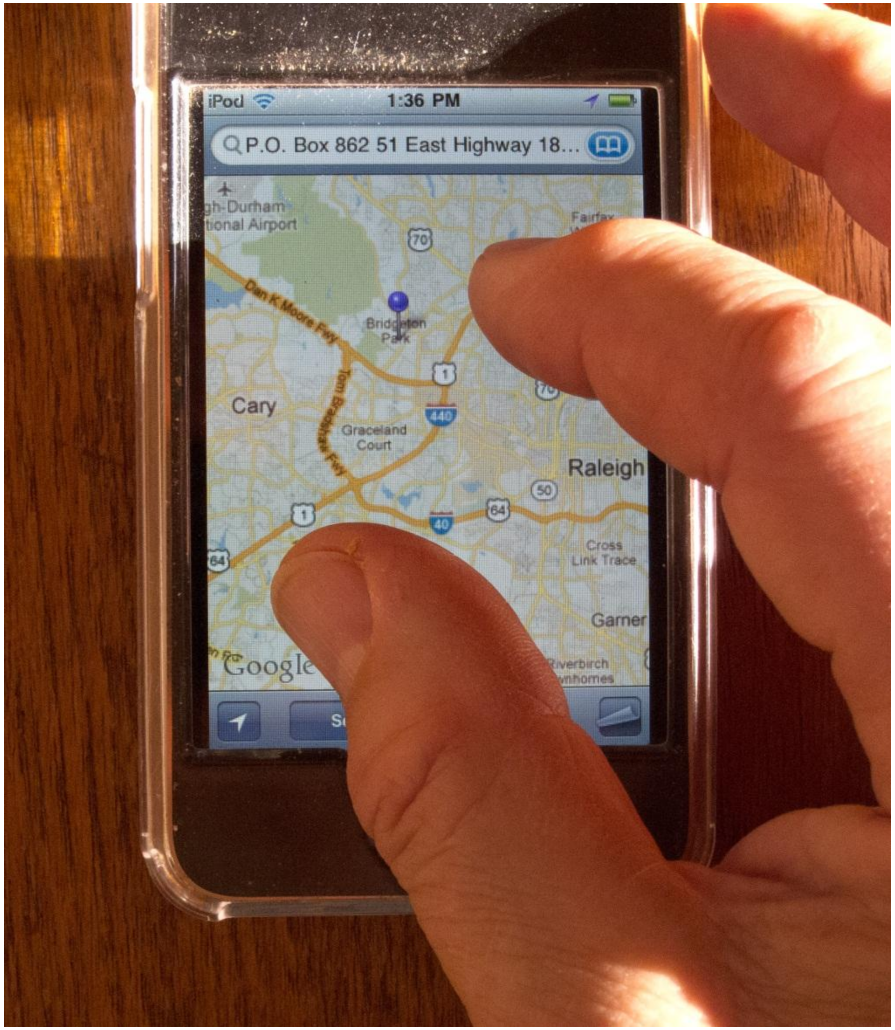
Reducing load: zooming in on map image (3)

Direct Manipulation



Reducing load: zooming in on map image (4)

Embodied interaction



Human Factors / Ergonomics Perspective

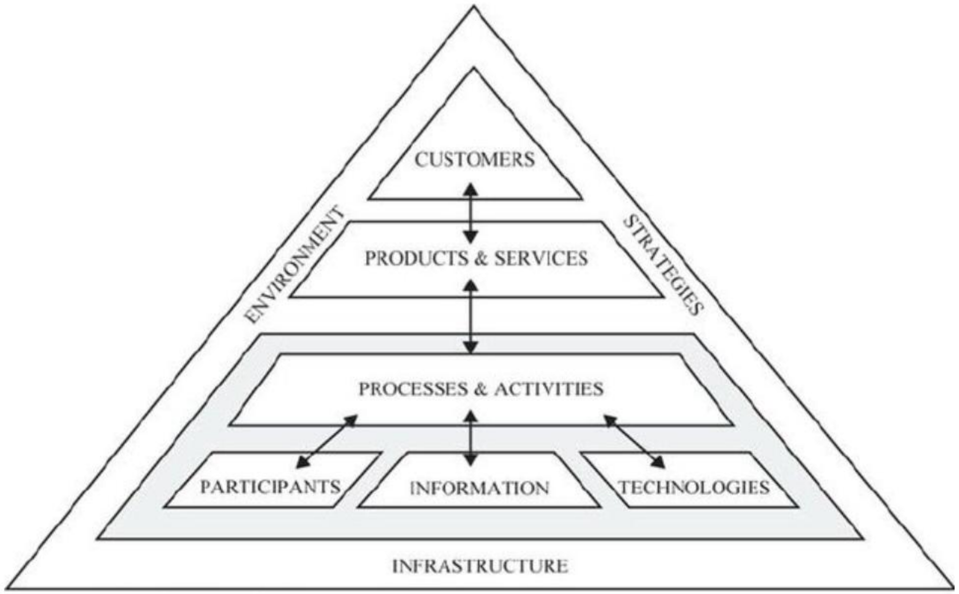
- What Is “Unnecessary Variability”?
  - Variability is natural in work systems
    - Demand changes
    - Task complexity varies
    - Human performance fluctuates
  - Necessary variability
    - Expert judgment adapting to context
    - Customization for customer needs
  - Unnecessary variability
    - Inconsistent methods due to lack of standards
    - Errors from ambiguous instructions
    - Quality fluctuation from unstable processes
    - Delays from unclear roles
  - Goal:
    - Reduce non-value-adding variability

Operations / Quality Perspective

- Work as transformation process:
  - Inputs → structured activities → outputs
- ISO process concept:
  - Interrelated activities
  - Transform inputs into outputs
- Enables analysis of:
  - Bottlenecks
  - Waiting
  - Variability
  - Capacity
- Focus:
  - System-level performance

Information Systems Perspective

- Common mistake:
  - Treating “system” as just IT
- Work system thinking:
  - Proper unit of analysis = work system
  - Includes:
    - People
    - Processes
    - Information
    - Technology
    - Customers



Alter’s work system framework

Information Systems Perspective

- Why IT projects fail:
  - Software works
  - But:
    - Roles
    - Procedures
    - Incentives
    - Training
    - Metrics

    → Not redesigned

User (Worker) Experience Perspective

- Definition of user experience (widely used in HCI/UX):
  - "User experience is a person's perceptions and responses resulting from the use and/or anticipated use of a product, system, or service (ISO 9241-210)."
  - "User experience is the totality of the internal effects experienced by a user as a result of interacting with a product, system, or service (Hassenzahl)"

User (Worker) Experience Perspective

- Core Components of User Experience (UX)
  - UX is often described as a mix of pragmatic and experiential dimensions (Hassenzahl's pragmatic vs. hedonic UX model):
    - Utility (Functionality)
      - Does it provide the capabilities users need to achieve goals?
    - Usability (Ease of Use)
      - How easily and reliably can users accomplish tasks?
    - Affects (Emotions & Feelings)
      - Emotional responses during/after use (e.g., trust, frustration, enjoyment)
    - Meaningfulness (Hedonic / Reflective Value)
      - Personal significance and longer-term value (e.g., identity, attachment)

User (Worker) Experience Perspective

- Meaningfulness

User (Worker) Experience Perspective

- User Experience: Work as Lived Interaction
  - Work is not only what is done, but how it is experienced
  - UX treats work as goal-directed activity situated in a work domain (context)
  - Users engage in sensory, cognitive, and physical actions
  - Work includes employment, learning, coordination, and even play (when play is goal-directed)

User (Worker) Experience Perspective

- Work Practice, Activity, and Domain
  - Work practice → How people actually perform work
    - Routines, conventions, skills, decision patterns
  - Work activity → Sensory, cognitive, and physical actions
  - Work domain → Broader context
    - Constraints
    - Workflows
    - Collaboration structures
    - Tools and rules

User (Worker) Experience Perspective

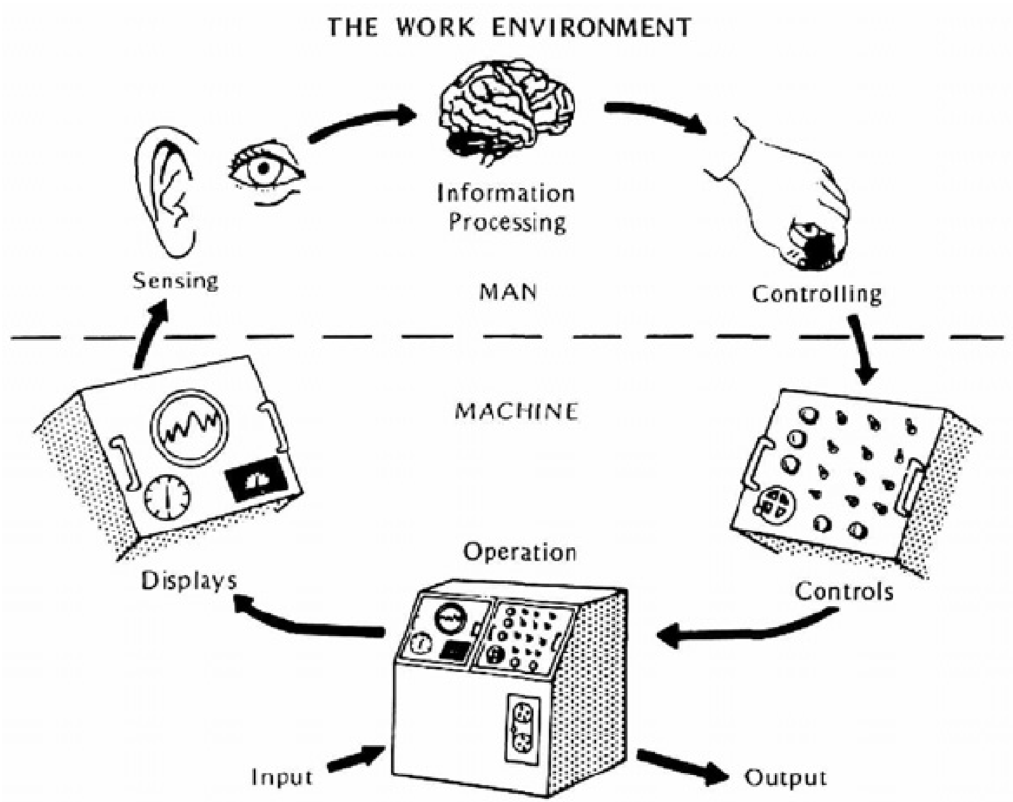
- Interface and Interaction (Core Definitions)
  - A system is an interaction space, not just an artifact
  - Interface
    - The boundary through which users perceive and act on the system
    - Includes displays (what users see) and controls (what users manipulate)
    - Can be physical or digital
  - Interaction
    - The dynamic exchange between user and system over time
    - User actions → system responses
    - Perception → decision → action → feedback (a control loop)

Interface = boundary

Interaction = exchange across the boundary

User (Worker) Experience Perspective

- Human-machine interaction



User (Worker) Experience Perspective

- Interaction Design as Coupling Architecture
  - Interaction design determines how components connect in practice. It structures:
    - Information access (visibility of system state)
    - Task sequencing (workflow logic)
    - Control design and input mappings
    - Feedback loops (confirmation, progress, quality signals)
    - Error detection and recovery
    - Authority and decision allocation

Interface defines the boundary.

Interaction defines the exchange.

Interaction design defines how work is enacted through that exchange.

Integrated Definition of Work

- Working definition for this course:
  - “Work is goal-directed activity in which humans and/or machines use resources (information, technology, materials, energy, time, space) to carry out tasks and processes through structured interactions mediated by interfaces, within social and organizational structures, producing valuable outputs (products, services, decisions, knowledge, experiences) for stakeholders.”

Integrated Definition of Work

- Work = goal-directed activity
- Performed by:
  - Humans and/or machines
- Embedded in social and organizational structures
  - Roles and responsibilities
  - Norms and rules
  - Authority and coordination
  - Division of labor
- Enacted through interactions
  - Structured exchanges between agents
- Mediated by interfaces
  - Human–machine, human–human, or machine–machine boundaries
- Using resources:
  - Information · Technology · Materials · Energy · Time · Space
- To carry out tasks and processes
- Producing valuable outputs:
  - Products · Services · Decisions · Knowledge · Experiences
- For stakeholders

Mini-Summary

- Work can be understood at multiple levels:
  - Behavior
  - Task
  - Process
  - Work system
- Across disciplines, a common conclusion:
  - Work is organized activity
  - It produces value
  - It operates through socio-technical arrangements